

Answer Sheet for Worksheet of Lecture 8

Answer 1: IoU Calculation

Given the bounding boxes:

- Predicted box: $(x, y, w, h) = (1, 1, 3, 6)$.
- Ground truth box: $(\hat{x}, \hat{y}, \hat{w}, \hat{h}) = (2, 2, 5, 5)$.

To calculate the Intersection over Union (IoU), we follow these steps:

1. Intersection Area:

Left-bottom: (2, 2)

Right-bottom: (4, 2)

Left-top: (2, 7)

Right-top: (4, 7)

The intersection area is:

$$\text{Intersection Area} = (4 - 2) \times (7 - 2) = 2 \times 5 = 10$$

2. Union Area:

- Area of predicted box $= 3 \times 6 = 18$
- Area of ground truth box $= 5 \times 5 = 25$
- Union Area $= 18 + 25 - 10 = 33$

3. IoU:

$$\text{IoU} = \frac{\text{Intersection Area}}{\text{Union Area}} = \frac{10}{33} \approx 0.3030$$

Answer 2: Bounding Box Loss Calculation

Given the following:

- Predicted bounding box: $(x_1, y_1, x_2, y_2) = (2, 2, 5, 5)$.
- Predicted class probabilities: $[0.8, 0.2]$ for class 1 and class 2.
- Ground truth bounding box: $(x_1, y_1, x_2, y_2) = (1, 1, 4, 4)$.
- Ground truth class label: class 1.

a) $L1$ and L_{iou} loss:

$L1$ loss for bounding boxes is calculated by taking the sum of the absolute differences between the predicted and ground truth bounding box coordinates:

$$L1 = |1 - 2| + |1 - 2| + |4 - 5| + |4 - 5| = 1 + 1 + 1 + 1 = 4$$

L_{iou} loss is calculated based on the Intersection over Union (IoU) between the predicted and ground truth bounding boxes:

$$L_{iou} = 1 - \frac{\text{Intersection Area}}{\text{Union Area}} = 1 - \frac{4}{9 + 9 - 4} \approx 0.7143$$

b) Cross-entropy loss for classification:

Cross-entropy loss (with natural log) for class 1 (ground truth) is calculated using the predicted probability for class 1:

$$L_{CE} = -\ln(0.8) \approx 0.2231$$

Thus, the total loss is the sum of the bounding box loss and classification loss:

$$L_{\text{total}} = \lambda_{L1}L1 + \lambda_{iou}L_{iou} + L_{CE} = 4 + 0.7143 + 0.2231 = 4.9374$$

Answer 3: Hungarian Matching

Given the following cost matrix for 4 predicted bounding boxes (columns) and 4 ground truth bounding boxes (rows):

0.1	0.5	0.2	0.7
0.2	0.6	0.3	0.1
0.4	0.3	0.5	0.6
0.8	0.2	0.4	0.3

We apply the Hungarian algorithm to find the optimal matching and compute the total cost.

Step 1: Subtract Row Minima

We subtract the minimum value from each row:

0	0.4	0.1	0.6
0.1	0.5	0.2	0
0.1	0	0.2	0.3
0.6	0	0.2	0.1

Step 2: Subtract Column Minima

We subtract the minimum value from each column:

0	0.4	0	0.6
0.1	0.5	0.1	0
0.1	0	0.1	0.3
0.6	0	0.1	0.1

Step 3: Cover the Zeros

We cover the zeros with the fewest number of straight lines. In this case, we cover the first and second rows, as well as the second column.

0	0.4	0	0.6
0.1	0.5	0.1	0
0.1	0	0.1	0.3
0.6	0	0.1	0.1

Step 4: Modify the Matrix

The smallest uncovered element is 0.1. After subtracting 0.1 from the uncovered elements and adding it to the double-covered elements, the updated matrix is:

0	0.5	0	0.6
0.1	0.6	0.1	0
0	0	0	0.2
0.5	0	0	0

Return to Step 3

Now the fewest number of straight lines is equal to 4, we can get the final matching result.

Final Matching and Total Cost

Matching result (other combinations exist where each row and column has exactly one 0):

- Ground Truth 1 (row 1) matches with Predicted Box 1 (column 1): Cost = 0.1
- Ground Truth 2 (row 2) matches with Predicted Box 4 (column 4): Cost = 0.1
- Ground Truth 3 (row 3) matches with Predicted Box 2 (column 3): Cost = 0.3
- Ground Truth 4 (row 4) matches with Predicted Box 3 (column 2): Cost = 0.4

Total Cost = $0.1 + 0.1 + 0.3 + 0.4 = 0.9$.

Thus, the optimal total cost of the matching is 0.9.

Answer 4: Max Pooling Operation

Given the following 5x5 matrix:

9	21	23	3	20
1	22	14	4	8
17	19	5	12	10
15	7	6	25	24
16	11	2	13	18

Perform max pooling with a 3x3 kernel, stride=2, and padding=1. To perform max pooling, we slide a 3x3 window over the padded matrix (with padding=1).

Padded matrix:

0	0	0	0	0	0	0
0	9	21	23	3	20	0
0	1	22	14	4	8	0
0	17	19	5	12	10	0
0	15	7	6	25	24	0
0	16	11	2	13	18	0
0	0	0	0	0	0	0

The result after max pooling (using 3x3 kernel, stride=2, padding=1):

22	23	20
22	25	25
16	25	25

Answer 5: Deconvolution Operation

Given the following 2x2 matrix:

1	2
3	4

and a 3x3 convolution kernel:

0	1	0
1	1	1
0	1	0

We perform a deconvolution operation (transpose convolution) with stride = 2 and padding = 1.

Step 1: Padding the input matrix with 1 zero between the original elements and 1 zero around the border:

0	0	0	0	0
0	1	0	2	0
0	0	0	0	0
0	3	0	4	0
0	0	0	0	0

Step 2: Perform transposed convolution (deconvolution) using the 3x3 kernel and stride=1:

1	3	2
4	0	6
3	7	4